BEE 4750 Homework 4: Linear Programming and Capacity Expansion

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Due Date

Thursday, 11/06/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to analytically (or graphically) solve a linear program.
- Problem 2 asks you to formulate and solve a resource allocation problem using linear programming.
- Problem 3 asks you to formulate, solve, and analyze a standard generating capacity expansion problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/Desktop/Cornell_Courses/FA_25/BEE_4750/hw4-xindilcindy`
   Installed BenchmarkTools  v1.6.3
   Installed ArgCheck        v2.5.0
   Installed MarkdownTables  v1.1.0
   Installed DisplayAs       v0.1.6
   Installed StatsBase       v0.34.7
   Installed DataFrames      v1.8.1
   Installed PrettyTables    v3.1.0
   Installed JuMP            v1.29.2
Precompiling project...
    470.8 ms    ArgCheck
    472.7 ms    StructUtils → StructUtilsTablesExt
    561.0 ms    DisplayAs
    876.8 ms    BenchmarkTools
   1557.7 ms    StatsBase
   1086.1 ms    MarkdownTables
  17491.5 ms    PrettyTables
  24067.6 ms    MathOptInterface
   1869.5 ms    MathOptIIS
   7976.6 ms    HiGHS
  14594.4 ms    JuMP
  39848.1 ms    Plots
  28463.4 ms    DataFrames
   1906.2 ms    Latexify → DataFramesExt
  14 dependencies successfully precompiled in 49 seconds. 208 already precompiled.
```

```julia
using JuMP
using HiGHS
using DataFrames
using Plots
using Measures
using CSV
using MarkdownTables
```

Problems (Total: 30 Points)

Problem 1 (Total: 3 Points)

Solve the following linear program **analytically**. Show your work and justify any reasoning.

$$
\begin{aligned}
\max_{x,y} \quad & 3x + 2y \\
\text{subject to:} \quad & x + 4y \leq 16 \\
& x - 2y \leq -1 \\
& 0 \leq x \leq 4 \\
& 1 \leq y \leq 4.
\end{aligned}
$$

Answer:

Rewrite the constraints as:

$$
\begin{aligned}
x &\leq 16 - 4y, \\
x &\leq 2y - 1, \\
0 \leq x &\leq 4, \\
1 \leq y &\leq 4.
\end{aligned}
$$

Thus, for each $y \in [1, 4]$,

$$0 \leq x \leq \min\{4,\, 16 - 4y,\, 2y - 1\}.$$

Compare the three upper bounds:

$$16 - 4y = 4 \Rightarrow y = 3, \quad 2y - 1 = 4 \Rightarrow y = 2.5.$$

Hence, by intervals of y:

$$
\begin{cases}
1 \leq y \leq 2.5 & \Rightarrow x \leq 2y - 1, \\
2.5 \leq y \leq 3 & \Rightarrow x \leq 4, \\
3 \leq y \leq 4 & \Rightarrow x \leq 16 - 4y.
\end{cases}
$$

The feasible region's vertices occur at intersections of these boundaries:

$$(x, y) = (0, 1), \quad (1, 1), \quad (4, 2.5), \quad (4, 3), \quad (0, 4).$$

Evaluate the objective function:

$$f(x, y) = 3x + 2y$$

(x,y)	$f(x,y)$
$(0,1)$	$3(0)+2(1)=2$
$(1,1)$	$3(1)+2(1)=5$
$(4,2.5)$	$3(4)+2(2.5)=17$
$(4,3)$	$3(4)+2(3)=18$
$(0,4)$	$3(0)+2(4)=8$

Conclusion:

The maximum value is attained at

$$(x^*, y^*) = (4,3),$$

with

$$f^* = 18.$$

At this optimum, the constraints $x = 4$ and $x + 4y = 16$ are active.

Problem 2 (12 points)

A farmer has access to a pesticide which can be used on corn, soybeans, and wheat fields, and costs $70/kg-yr to apply. The crop yields the farmer can obtain following crop yields by applying varying rates of pesticides to the field are shown in Table 1.

Application Rate (kg/ha)	Soybean (kg/ha)	Wheat (kg/ha)	Corn (kg/ha)
0	2900	3500	5900
1	3800	4100	6700
2	4400	4200	7900

Table 1: Crop yields from applying varying pesticide rates for Problem 1.

The costs of production, *excluding pesticides*, for each crop, and selling prices, are shown in Table 2.

Crop	Production Cost ($/ha-yr)	Selling Price ($/kg)
Soybeans	350	0.36
Wheat	280	0.27
Corn	390	0.22

Table 2: Costs of crop production, excluding pesticides, and selling prices for each crop.

Recently, environmental authorities have declared that farms cannot have an *average* application rate on soybeans, wheat, and corn which exceeds 0.8, 0.7, and 0.6 kg/ha, respectively. The farmer has asked you for advice on how they should plant crops and apply pesticides to maximize profits over 130 total ha while remaining in regulatory compliance if demand for each crop (which is the maximum the market would buy) this year is 250,000 kg?

Problem 2.1

Formulate a linear program for this resource allocation problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations). **Tip: Make sure that all of your constraints are linear**.

Answer:

Sets and indices: - $c \in \{S, W, C\}$: crops (Soybean, Wheat, Corn).
- $r \in \{0, 1, 2\}$: pesticide application rates (kg/ha).

Parameters: - $A_{\max} = 130$ ha - total available area.
- $D_c = 250{,}000$ kg - maximum demand for crop c.
- $y_{c,r}$ (kg/ha) - yield of crop c at rate r (from Table 1).
- p_c (\$/kg) - selling price for crop c.
- g_c (\$/ha) - production cost for crop c excluding pesticide.
- L_c (kg/ha) - regulatory maximum average application for crop c ($L_S = 0.8$, $L_W = 0.7$, $L_C = 0.6$).
- Pesticide cost = \$70 per kg applied (cost per ha at rate r is $70r$).

Decision variables:

For each crop c and rate r:

$$x_{c,r} = \text{area (ha) planted to crop } c \text{ with pesticide rate } r, \quad x_{c,r} \geq 0.$$

Objective function:

Profit per hectare for crop c at rate r:

$$\pi_{c,r} = p_c \, y_{c,r} - g_c - 70r \quad (\$/\text{ha})$$

Total profit to maximize:

$$\max Z = \sum_{c \in \{S,W,C\}} \sum_{r \in \{0,1,2\}} \pi_{c,r} \, x_{c,r}.$$

Constraints: 1. Total area limit:

$$\sum_c \sum_r x_{c,r} \leq A_{\max} = 130.$$

2. Per-crop demand limits (cannot produce more than market will buy):

$$\sum_r y_{c,r}\, x_{c,r} \le D_c, \quad \forall c.$$

3. Regulatory average-application limits: The average application for crop c is $\dfrac{\sum_r r\, x_{c,r}}{\sum_r x_{c,r}} \le L_c$. Multiplying both sides by $\sum_r x_{c,r}$ gives a linear constraint:

$$\sum_r r\, x_{c,r} \le L_c \sum_r x_{c,r}, \quad \forall c,$$

or equivalently,

$$\sum_r (r - L_c)\, x_{c,r} \le 0, \quad \forall c.$$

4. Nonnegativity:

$$x_{c,r} \ge 0, \quad \forall c, r.$$

Problem 2.2

How many ha should the farmer dedicate to each crop and with what pesticide application rate(s)? How much profit will the farmer expect to make?

```
# Sets & indices
crops = [:S, :W, :C]           # Soybean, Wheat, Corn
rates = [0, 1, 2]              # Pesticide application rates (kg/ha)

# Parameters
Amax = 130.0                   # total land area (ha)
D = Dict(c => 250_000.0 for c in crops)  # demand cap (kg)
L = Dict(:S => 0.8, :W => 0.7, :C => 0.6)  # regulatory max avg pesticide (kg/ha)

# Yields (kg/ha)
y = Dict(
    :S => Dict(0 => 2900, 1 => 3800, 2 => 4400),
    :W => Dict(0 => 3500, 1 => 4100, 2 => 4200),
    :C => Dict(0 => 5900, 1 => 6700, 2 => 7900)
)

# Selling prices ($/kg)
p = Dict(:S => 0.36, :W => 0.27, :C => 0.22)

# Production costs excluding pesticide ($/ha)
```

```
g = Dict(:S => 350, :W => 280, :C => 390)

# Pesticide cost ($/kg)
pesticide_cost = 70.0

# Model
model = Model(HiGHS.Optimizer)
set_silent(model)

# Decision variables: area planted to crop c with rate r (ha)
@variable(model, x[c in crops, r in rates] >= 0)

# Objective: maximize total profit
@objective(model, Max,
    sum((p[c] * y[c][r] - g[c] - pesticide_cost * r) * x[c, r]
        for c in crops, r in rates)
)

# Constraints

# 1. Total area limit
@constraint(model, land_constraint, sum(x[c, r] for c in crops, r in rates) <= Amax)

# 2. Per-crop demand (market cap)
@constraint(model, [c in crops],
    sum(y[c][r] * x[c, r] for r in rates) <= D[c])

# 3. Regulatory average-application limits (linearized form)
@constraint(model, [c in crops],
    sum((r - L[c]) * x[c, r] for r in rates) <= 0)

# Solve
optimize!(model)

# Results
println("\nSolver status: ", termination_status(model))
println("Optimal profit: \$", objective_value(model))

println("\nOptimal area allocation (ha):")
for c in crops
    for r in rates
        println("  Crop $(c), rate $(r) kg/ha: ", round(value(x[c, r]); digits = 2))
```

```
    end
end
```

```
Solver status: OPTIMAL
Optimal profit: $116741.16702082448

Optimal area allocation (ha):
  Crop S, rate 0 kg/ha: 13.81
  Crop S, rate 1 kg/ha: 55.25
  Crop S, rate 2 kg/ha: 0.0
  Crop W, rate 0 kg/ha: 6.74
  Crop W, rate 1 kg/ha: 15.73
  Crop W, rate 2 kg/ha: 0.0
  Crop C, rate 0 kg/ha: 26.92
  Crop C, rate 1 kg/ha: 0.0
  Crop C, rate 2 kg/ha: 11.54
```

Answer:

Allocation of crop areas:

- Soybean — 1 kg/ha: 55.25 ha and 0 kg/ha: 13.81 ha (total soybean 69.06 ha)
- Wheat — 1 kg/ha: 15.73 ha and 0 kg/ha: 6.74 ha (total wheat 22.48 ha)
- Corn — 0 kg/ha: 26.92 ha, 2 kg/ha: 11.54 ha (total corn 38.46 ha)

Expected profit: $116,741.17

Problem 2.3

The farmer has an opportunity to buy an extra 10 ha of land. How much extra profit would this land be worth to the farmer? Discuss why this value makes sense and under what conditions you would recommend the farmer should make the purchase.

```
# Shadow price of the land constraint
shadow_price = -dual(land_constraint)
println("\nShadow price (value of 1 extra ha): \$", shadow_price)

# Value of 10 additional hectares
println("Value of 10 extra ha: \$", 10 * shadow_price)
```

```
Shadow price (value of 1 extra ha): $729.4
Value of 10 extra ha: $7294.0
```

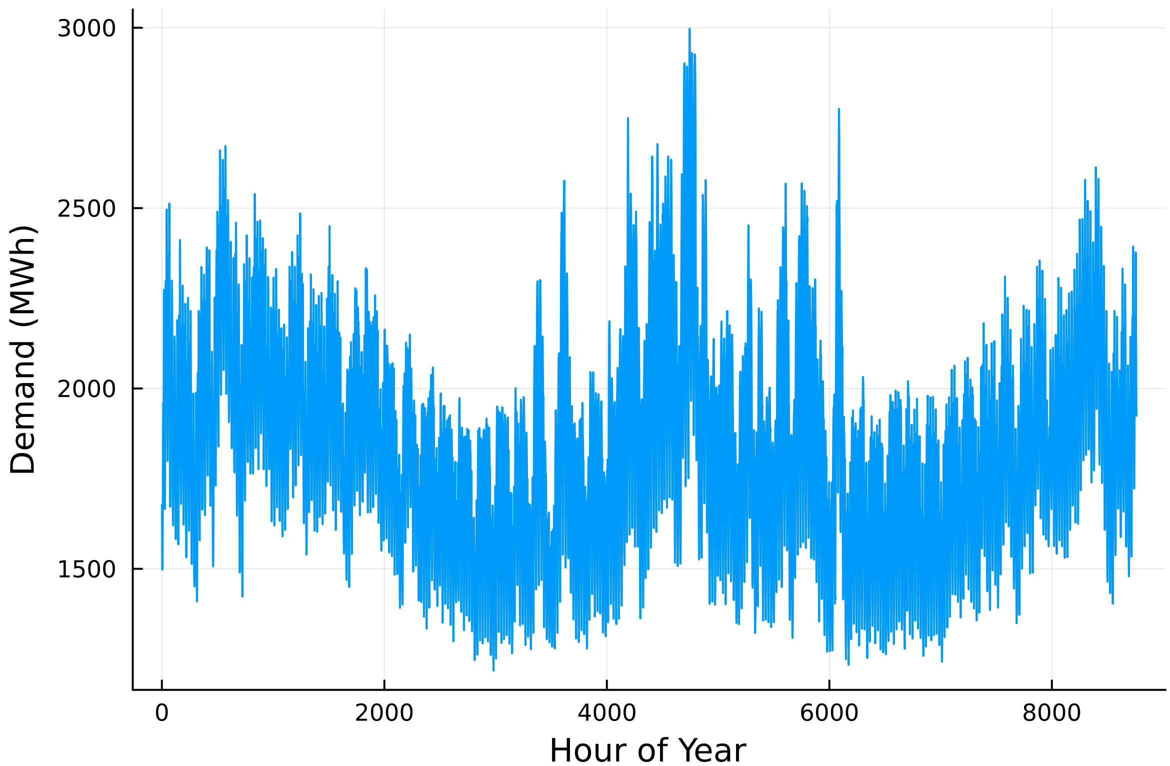
The marginal value, or shadow price, of land from the LP is approximately \$729.4 per hectare per year, so 10 ha are worth about \$7,294 in additional annual profit. I recommend purchasing the 10 ha if and only if the annualized cost of the land is less than \$7,294, or less than \$729/ha, thus the farmer can make profit this way. Otherwise, the effect will be opposite and the loss outweighs the gains. Before buying, the farmer should account for financing costs, transaction costs, taxes, and risk. The farmer may also needs to resolve the LP under the larger land area to verify the marginal value remains similar.

Problem 3 (15 points)

For this problem, we will use hourly load (demand) data from 2013 in New York's Zone C (which includes Ithaca). The load data is loaded and plotted below in Figure 1.

```julia
# load the data, pull Zone C, and reformat the DataFrame
NY_demand = DataFrame(CSV.File("data/2013_hourly_load_NY.csv"))
rename!(NY_demand, :"Time Stamp" => :Date)
demand = NY_demand[:, [:Date, :C]]
rename!(demand, :C => :Demand)
demand[:, :Hour] = 1:nrow(demand)

# plot demand
plot(demand.Hour, demand.Demand, xlabel = "Hour of Year", ylabel = "Demand (MWh)", label = :
```



Next, we load the generator data, shown in Table 3. This data includes fixed costs ($/MW installed), variable costs ($/MWh generated), and CO2 emissions intensity (tCO2/MWh generated).

```
gens = DataFrame(CSV.File("data/generators.csv"))
```

	Plant	FixedCost	VarCost	Emissions
	String15	Int64	Int64	Float64
1	Geothermal	450000	0	0.0
2	Coal	220000	24	1.0
3	NG CCGT	82000	30	0.43
4	NG CT	65000	40	0.55
5	Wind	91000	0	0.0
6	Solar	70000	0	0.0

Finally, we load the hourly solar and wind capacity factors, which are plotted in Figure 2. These tell us the fraction of installed capacity which is expected to be available in a given hour for generation (typically based on the average meteorology).

```
# load capacify factors into a DataFrame
cap_factor = DataFrame(CSV.File("data/wind_solar_capacity_factors.csv"))

# plot January capacity factors
p1 = plot(cap_factor.Wind[1:(24*31)], label = "Wind")
plot!(cap_factor.Solar[1:(24*31)], label = "Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

p2 = plot(cap_factor.Wind[4344:4344+(24*31)], label = "Wind")
plot!(cap_factor.Solar[4344:4344+(24*31)], label = "Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

display(p1)
display(p2)
```

You have been asked to develop a generating capacity expansion plan for the utility in Riley County, NY, which currently has no existing electrical generation infrastructure. The utility can build any of the following plant types: geothermal, coal, natural gas combined cycle gas turbine (CCGT), natural gas combustion turbine (CT), solar, and wind. The NY state legislature is planning to enact an annual CO2 limit, which for the utility would limit the emissions in its footprint to 1.5 MtCO2/yr.

While coal, CCGT, and CT plants can generate at their full installed capacity, geothermal plants operate at maximum 85% capacity, and solar and wind available capacities vary by the hour depend on the expected meteorology. The utility will also penalize any non-served demand at a rate of $10,000/MWh.

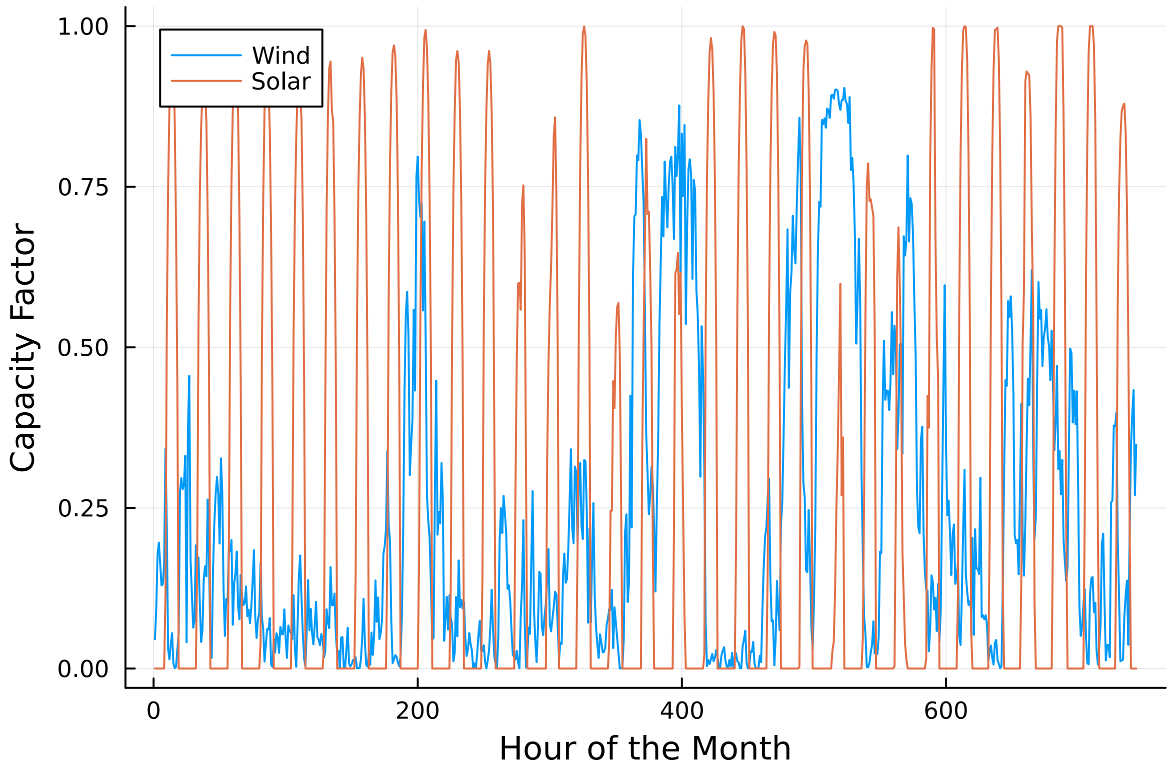
Problem 3.1

Formulate a linear program for this capacity expansion problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations).

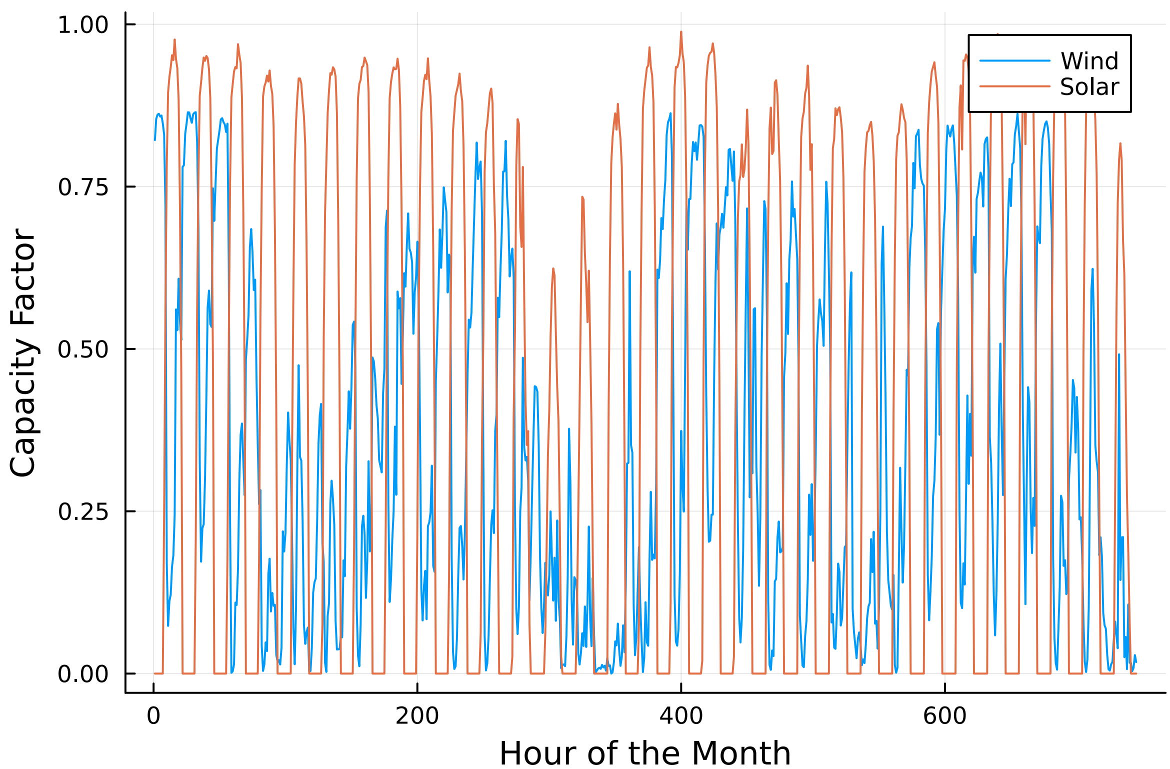
Answer:

Decision variables (all 0): - K_p = installed capacity of plant type p (MW) - $g_{p,t}$ = generation from plant p in hour t (MWh) - u_t = unmet demand in hour t (MWh)

Parameters: - D_t = electricity demand in hour t (MWh) - Fixed_p = annual fixed cost of capacity for plant p ($/MW\text{-}yr$) - Var_p = variable generation cost ($/MWh$) - Em_p = emissions intensity (tCO_2/MWh) - For coal, NG CCGT, NG CT: $\text{cf}_{p,t} = 1.0$ - For geothermal: $\text{cf}_{\text{geo},t} = 0.85$ - For



Capacity Factor
1.00
0.75
0.50
0.25
0.00
0
200
400
600
Hour of the Month
Wind
Solar



Capacity Factor
1.00
0.75
0.50
0.25
0.00
0
200
400
600
Hour of the Month
Wind
Solar

wind and solar: $\text{cf}_{p,t}$ is given by hourly data - VOLL = 10,000 $/MWh$ - CO2limit = 1.5×10^6 tCO_2/yr

Objective function (minimize total cost):

$$\min \sum_p \text{Fixed}_p K_p + \sum_t \sum_p \text{Var}_p g_{p,t} + \sum_t \text{VOLL}\, u_t$$

Demand balance (every hour t):

$$\sum_p g_{p,t} + u_t = D_t$$

Capacity limits:

For coal, NG CCGT, NG CT:

$$g_{p,t} \leq K_p$$

For geothermal:

$$g_{geo,t} \leq 0.85\, K_{geo}$$

For wind and solar:

$$g_{p,t} \leq \text{cf}_{p,t}\, K_p$$

Annual emissions constraint:

$$\sum_t \sum_p \text{Em}_p\, g_{p,t} \leq \text{CO2limit}$$

Nonnegativity:

$$K_p \geq 0, \quad g_{p,t} \geq 0, \quad u_t \geq 0$$

Problem 3.2

How much should the utility build of each type of generating plant? What will the total cost be? How much energy will be non-served?

```
# Hours and demand vector
T = nrow(demand)
D = demand.Demand

# Extract generator sets
plants = gens.Plant

# Turn generator columns into dictionaries
Fixed = Dict(gens.Plant .=> gens.FixedCost)
```

```julia
Var    = Dict(gens.Plant .=> gens.VarCost)
Em     = Dict(gens.Plant .=> gens.Emissions)

# Build capacity factor dictionary
cf = Dict(
    "Coal"       => fill(1.0, T),
    "NG CCGT"    => fill(1.0, T),
    "NG CT"      => fill(1.0, T),
    "Geothermal" => fill(0.85, T),
    "Wind"       => cap_factor.Wind,
    "Solar"      => cap_factor.Solar
)

VOLL = 10_000
CO2limit = 1.5e6

# Optimization model
model = Model(HiGHS.Optimizer)
set_silent(model)

@variable(model, K[p in plants] >= 0)             # MW capacity to build
@variable(model, g[p in plants, t=1:T] >= 0)     # hourly generation
@variable(model, u[t=1:T] >= 0)                  # unmet demand

# Objective: fixed + variable + unmet load penalty
@objective(model, Min,
    sum(Fixed[p] * K[p] for p in plants) +
    sum(Var[p] * g[p,t] for p in plants for t in 1:T) +
    sum(VOLL * u[t] for t in 1:T)
)

# Demand balance
@constraint(model, [t=1:T], sum(g[p,t] for p in plants) + u[t] == D[t])

# Capacity limits
@constraint(model, [p in plants, t = 1:T], g[p,t] <= cf[p][t] * K[p])

# CO  cap
co2_con = @constraint(model, sum(Em[p] * g[p,t] for p in plants for t in 1:T) <= CO2limit)

optimize!(model)
```

```julia
# Results
println("\nOptimal Capacity Build (MW):")
for p in plants
    println(p, ": ", round(value(K[p]), digits = 2))
end

println("\nTotal Annual Cost (\$): ", round(objective_value(model), digits = 2))
println("\nTotal Unserved Energy (MWh): ", round(sum(value.(u)), digits = 2))
```

```
Optimal Capacity Build (MW):
Geothermal: 285.57
Coal: 0.0
NG CCGT: 1509.17
NG CT: 703.06
Wind: 2548.89
Solar: 2103.75

Total Annual Cost ($): 7.8535295894e8

Total Unserved Energy (MWh): 332.29
```

Answer:

From the solution, it shows that larger capacity should be placed in renewables, wind and solar, and it also relies natural-gas-fired plants, primarily CCGT, for dispatchable firm energy and flexibility. Geothermal provides a modest baseload contribution. Coal is not built, indicating it is not cost-effective given the operating costs and the CO_2 cap. The very small amount of unserved energy shows that the system meets demand almost everywhere, which the model tolerates as well.

In summary, under the stated assumptions, the utility should build roughly 2.55 GW of wind, 2.10 GW of solar, 1.51 GW of NG CCGT, 0.70 GW of NG CT, and 0.29 GW of geothermal. This yields a total annual cost of about 7.85×10^8 and only 332.3 MWh of non-served energy, indicating a low-cost, reliable system consistent with the demand and generator data.

Problem 3.3

What fraction of annual generation does each plant type produce? How does this compare to the breakdown of built capacity that you found in Problem 3.2? Do these results make sense given the demand and generator data?

```julia
# Calculate Total Annual Generation by Plant
gen_by_plant = Dict(p => sum(value.(g[p, t]) for t in 1:length(demand.Hour)) for p in plants)

# Total Generation
total_gen = sum(values(gen_by_plant))

# Fraction of Annual Generation
gen_share = Dict(p => gen_by_plant[p] / total_gen for p in plants)

println("\nAnnual Generation (MWh):")
for p in plants
    println("  $(p): ", round(gen_by_plant[p]; digits = 2))
end

println("\nGeneration Share (% of total):")
for p in plants
    println("  $(p): ", round(100 * gen_share[p]; digits = 2), "%")
end

# Capacity Shares
cap_total = sum(value.(K[p]) for p in plants)
cap_share = Dict(p => value(K[p]) / cap_total for p in plants)

println("\nCapacity Share (% of total):")
for p in plants
    println("  $(p): ", round(100 * cap_share[p]; digits=2), "%")
end
```

```
Annual Generation (MWh):
  Geothermal: 1.09699546e6
  Coal: 0.0
  NG CCGT: 3.32276656e6
  NG CT: 129473.42
  Wind: 6.45174372e6
  Solar: 5.36659446e6

Generation Share (% of total):
  Geothermal: 6.7%
  Coal: 0.0%
  NG CCGT: 20.3%
  NG CT: 0.79%
```

```
  Wind: 39.42%
  Solar: 32.79%

Capacity Share (% of total):
  Geothermal: 3.99%
  Coal: 0.0%
  NG CCGT: 21.11%
  NG CT: 9.83%
  Wind: 35.65%
  Solar: 29.42%
```

Answer:

From the generation share and capacity share information - Wind and solar provide a larger share of generation compared to their shares of capacity, likely because their capacity factors during most hours are relatively favorable and they have zero variable cost, so the model uses them whenever available. - Geothermal generates 6.7% of total energy with only 4% of capacity. This may be attributed to the fact that geothermal is modeled with a constant 85% capacity factor, giving it a high utilization rate. - NG CCGT contributes 20.3% of energy, roughly equal to its 21.1% of capacity. This potentially indicates CCGT units operate as the main middle thermal plants, filling in gaps when renewables are not sufficient. - NG CT has 9.8% of capacity but produces less than 1% of energy. This aligns with its role as peaking plants, used only in high-demand or low-renewable hours due to high variable cost. - Coal has 0% capacity and 0% generation, meaning it is never economical under the cost and emissions constraints. The emissions cap strongly discourages coal.

Overall, the results are consistent with both the cost and operational characteristics of the system, as the generation shares align logically with the capacity mix as well as the economic and emissions assumptions in the problem.

Problem 3.4

What is the shadow price of CO2? What is the value to the utility be of allowing it to emit an additional 1000 tCO2/yr?

```
# Get shadow price (dual) of the CO  cap
shadow_co2 = -dual(co2_con)   # units: $ per tCO2

# Value of allowing +1000 tCO2/year
value_1000 = shadow_co2 * 1000

println("\nShadow price of CO  (dollars/tCO ): ", round(shadow_co2, digits = 4))
println("Value of allowing +1000 tCO /yr (dollars/yr): ", round(value_1000, digits = 2))
```

```
Shadow price of CO  (dollars/tCO ): 182.7719
Value of allowing +1000 tCO /yr (dollars/yr): 182771.94
```

Answer:

The shadow price of the CO emissions constraint is approximately $183 per tCO . This indicates that reducing the annual emissions limit by one tonne of CO increases the total system cost by about $183, while relaxing the constraint by one tonne decreases annual cost by the same amount. Therefore, allowing the utility to emit an additional 1000 tCO /yr would reduce total annual cost by roughly $182,772 per year The positive shadow price confirms that the CO constraint is binding, and that the system is already operating at the emissions limit.

> **Significant Digits**
>
> Use `round(x; digits=n)` to report values to the appropriate precision! If your number is on a different order of magnitude and you want to round to a certain number of significant digits, you can use `round(x; sigdigits=n)`.
>
> **Getting Variable Output Values**
>
> `value.(x)` will report the values of a `JuMP` variable `x`, but it will return a special container which holds other information about `x` that is useful for `JuMP`. This means that you can't use this output directly for further calculations. To just extract the values, use `value.(x).data`.
>
> **Suppressing Model Command Output**
>
> The output of specifying model components (variable or constraints) can be quite large for this problem because of the number of time periods. If you end a cell with an `@variable` or `@constraint` command, I *highly* recommend suppressing output by adding a semi-colon after the last command, or you might find that your notebook crashes.

References

List any external references consulted, including classmates.

Used ChatGPT for problems 2.2, 3.2, 3.3, mainly for helping understand how the coding language works and what questions are asking, not for discussions and analysis of problems. OpenAI. (2025). ChatGPT (Feb 1 GPT-4 model) [Large language model]. https://chat.openai.com